# LOCALIZING A RING ALONG A CLOSED PROFINITE SUBSET PRODUCES A W-LOCAL RING WITH THAT SET AS ITS CLOSED POINTS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the identification of the inverse image T with V(I) through the homeomorphism of profinite spaces, the construction of the localization along T with its w-locality and closed points, and the isomorphism of the quotients through faithfulness of local isomorphisms.

*Editorial note.* The label lemma-localize-along-closed-profinite names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-localize-along-closed-profinite –start 1011 –end 1031 –local-id 28005 returned RESULT=OK (L28005 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- The w-localization and its set of closed points. Same chapter, the construction of the w-localization A -> A_w with its closed point set Z mapping bijectively onto the spectrum of A. Supplies the w-localization A_w and the closed subset Z used to cut out the profinite subset.
- Homeomorphism of a bijective map of profinite spaces. Topology, the lemma that a continuous bijection from a profinite space to a compact Hausdorff space is a homeomorphism. Supplies the homeomorphism T -> V(I) for the inverse image T of V(I) in Z.
- The localization construction and faithfulness of local isomorphisms. Same chapter, the localization construction attached to a locally closed subscheme and the lemma that a local isomorphism which is a homeomorphism on spaces is an isomorphism. Supply the ring B = (A_w)_T^tilde and the conclusion that A/I -> B/IB is an isomorphism.

*Source reference.* Original reference 096L: sources/stacks-proetale/pinned-original-source.tex, source lines 603–603. Prerequisite pointer invoked at proof line(s) 1030; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 096V: sources/stacks-proetale/pinned-original-source.tex, source lines 710–710. Prerequisite pointer invoked at proof line(s) 1025; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0975: sources/stacks-proetale/pinned-original-source.tex, source lines 844–844. Prerequisite pointer invoked at proof line(s) 1021; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 08YE: sources/stacks-chow/tags.txt, source lines 11337–11337. Prerequisite pointer invoked at proof line(s) 1024; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $A$ be a ring. Let $V(I) \subset \mathrm{Spec}(A)$ be a closed subset which is a profinite topological space. Then there exists an ind-Zariski ring map $A \to B$ such that $\mathrm{Spec}(B)$ is w-local, the set of closed points is $V(IB)$, and $A/I \cong B/IB$.*

*Proof.* Let $A \to A_w$ and $Z \subset Y = \mathrm{Spec}(A_w)$ as in Lemma 0975. Let $T \subset Z$ be the inverse image of $V(I)$. Then $T \to V(I)$ is a homeomorphism by Topology, Lemma 08YE. Let $B = (A_w)^\sim_T$, see Lemma 096V. It is clear that $B$ is w-local with closed points $V(IB)$. The ring map $A/I \to B/IB$ is ind-Zariski and induces a homeomorphism on underlying topological spaces. Hence it is an isomorphism by Lemma 096L. $\square$